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Networking protocols used in devsecops

In **DevSecOps**, networking protocols are critical because they enable **secure communication, automation, monitoring, deployment, and compliance** across pipelines, cloud infrastructure, and production systems.

1. HTTP / HTTPS

Full Form: HyperText Transfer Protocol / Secure

What it does

- Transfers data between client and server
- HTTPS encrypts data using **TLS/SSL**

DevSecOps Usage

- REST APIs (CI/CD tools, cloud APIs)
- Web applications
- Kubernetes services
- Secure communication between microservices

Example

```
curl https://api.github.com
```

Security Aspect

- HTTPS prevents **MITM attacks**
- Certificates managed via **Let's Encrypt / ACM**

2. SSH

Full Form: Secure Shell

What it does

- Secure remote access to servers

DevSecOps Usage

- Access Linux servers
- Git operations (GitHub, GitLab)
- Configuration management (Ansible)
- Kubernetes node access

Example

```
ssh ec2-user@10.0.0.5
```

Security Best Practice

- Use **key-based authentication**
- Disable password login

3. FTP / SFTP

Full Form: File Transfer Protocol / Secure FTP

What it does

- Transfers files between systems

DevSecOps Usage

- Artifact transfer (legacy systems)
- Backup uploads

Security Note ⚠️

- FTP is **insecure**
- Always prefer **SFTP**

4. DNS

Full Form: Domain Name System

What it does

- Converts domain names to IP addresses

DevSecOps Usage

- Service discovery (Kubernetes, cloud)
- Load balancers
- External access to applications

Example

```
nslookup google.com
```

Security Concern

- DNS spoofing
- Use **DNSSEC**

5. TCP / UDP

TCP (Transmission Control Protocol)

- Reliable, ordered, connection-oriented
- Used by: **HTTP, HTTPS, SSH**

UDP (User Datagram Protocol)

- Fast, connectionless
- Used by: **DNS, monitoring, streaming**

DevSecOps Usage

- Application traffic
- Monitoring agents
- Logging pipelines

6. TLS / SSL

Full Form: Transport Layer Security

What it does

- Encrypts data in transit

DevSecOps Usage

- HTTPS
- Secure APIs
- Secure microservice communication
- Kubernetes Ingress TLS

Security Focus

- Certificate rotation
- Enforce TLS 1.2+

7. SMTP

Full Form: Simple Mail Transfer Protocol

What it does

- Sends emails

DevSecOps Usage

- CI/CD pipeline alerts
- Security notifications
- Incident reports

Example

- Jenkins build failure emails
- Security scan alerts

8. SNMP

Full Form: Simple Network Management Protocol

What it does

- Monitors network devices

DevSecOps Usage

- Infrastructure monitoring
- Network health checks

Security Note

- Use **SNMPv3** (encrypted)

9. ICMP

Full Form: Internet Control Message Protocol

What it does

- Network diagnostics

DevSecOps Usage

- Health checks
- Network troubleshooting

Example

```
ping google.com
```

10. NTP

Full Form: Network Time Protocol

What it does

- Synchronizes system time

DevSecOps Usage

- Log correlation
- Security audits
- Certificate validation

11. API Protocols (REST / gRPC)

REST

- HTTP-based APIs
- Used by most CI/CD tools

gRPC

- High-performance communication
- Used in microservices

DevSecOps Usage

- Jenkins ↔ GitHub
- Kubernetes APIs
- Cloud provider APIs

12. Webhooks

What it does

- Event-based communication

DevSecOps Usage

- Trigger pipelines
- Security scan triggers
- Auto deployments

Example

- Git push triggers Jenkins pipeline

13. Kubernetes Networking Protocols

- **CNI** – Container networking
- **Service Mesh (Istio)** – mTLS, traffic control
- **Ingress (HTTP/HTTPS)**

Security Features

- mTLS
- Network policies
- Zero Trust

Interview One-Line Summary

DevSecOps uses networking protocols to ensure secure, automated, reliable communication across CI/CD pipelines, cloud infrastructure, and production systems.

Quick Interview Table

Protocol	Purpose	DevSecOps Use
HTTPS	Secure web traffic	APIs, apps
SSH	Secure access	Server management
DNS	Name resolution	Service discovery
TLS	Encryption	Zero trust
TCP/UDP	Data transport	App traffic
SMTP	Alerts	CI/CD notifications
SNMP	Monitoring	Infra health
ICMP	Troubleshooting	Network checks